

Smart BI Onboarding & KPI Flow

System Validation, Stress-Testing, & UX Architecture Mitigations

1. Executive Summary

This document summarizes the system validation and stress-testing of the Smart BI conversational onboarding flow. We analyzed the proposed design logic against millions of potential business queries across SaaS, E-commerce, Healthcare, Finance, and Logistics sectors to verify its structural integrity for senior product design interviews.

2. Smart BI Flow Logic

Zero-Selection Onboarding: KPIs are auto-generated from data schemas (CSV, Excel, DB, API) without manual table joining or column mapping.

Flow A (Specific Queries): Direct metrics mapping + evaluation against dynamic, carried-forward targets (e.g. 'Revenue was 42L, which missed the 50L target by 16%').

Flow B (Ambiguous Queries): Multi-KPI ambiguity is resolved through exactly one clarifying question (e.g. 'What should I base this on? Revenue units or margin?').

3. Primary Failure Modes & Blind Spots

Multi-Dimensional Ambiguity: Vague queries like 'Are we on track?' contain metric, time, and regional ambiguity. A single binary question cannot resolve all three, returning incomplete data.

Compound Formulas: AI cannot auto-infer complex metrics (e.g., LTV:CAC, NPS, or Net Retention Rate) purely from column names without a semantic formula model.

Combinatorial Explosion: Injecting large enterprise databases with 10,000+ tables will generate a massive amount of garbage KPIs without curation rules.

Target seasonality mismatch: Auto-carrying targets forward breaks for seasonal business periods (e.g., Retail sales spikes in December).

4. Architectural Recommendations

Metric Store Graph: Map raw schemas to a semantic node graph containing pre-defined formulas (LTV, CAC, Retention) to resolve complex ratios automatically.

Progressive Clarification: Upgrade Flow B to ask compound clarification options (e.g. 'Revenue or Margin for this month or YTD?') when multi-dimensional parameters are missing.

Dimensional Target Bindings: Bind targets to specific scopes (Value, Metric, Timeframe, Segment) instead of flat company-wide values.

5. The Hybrid UX Model (Post-Onboarding Selection)

The design wireframes introduce a structured KPI Checklist & Selector panel for chart creation after onboarding, bridging conversational query systems with classic graphical user interfaces.

Recognition over Recall: The sidebar checklist allows users to search and select active KPIs visually, reducing cognitive load compared to memorizing exact database field names.

Seamless Disambiguation: Checkbox selections or conversational pills (e.g., Flow B) act as physical disambiguation triggers, mapping user intents directly to active database formulas.

Role-Based Curation: Linking the selector to role-based schemas (CEO, Sales, HR) limits visible KPIs to relevant variables, narrowing search scopes and preventing database clutter.

6. Conclusion & Verdict

The Smart BI onboarding flow is highly viable, particularly when supported by the hybrid selector checklist which mitigates open-ended conversational ambiguity. To operate at scale with compound metrics and seasonal targets, the system should bind targets to seasonal scopes and integrate a semantic formula graph layer.